

Smart Hiring: MERN Job Portal with AI-Driven ATS Resume Optimization

A PROJECT REPORT

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IN

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DATA SCIENCE**

At



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PRESIDENCY UNIVERSITY

SCHOOL OF COMPUTER SCIENCE ENGINEERING

CERTIFICATE

This is to certify that the Project report “**Smart Hiring: MERN Job Portal with AI-Driven ATS Resume Optimization**” being submitted by G Hemanth Kumar bearing roll number 20211CSD0073 in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering Data Science is a bonafide work carried out under my supervision.

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DECLARATION

We hereby declare that the work, which is being presented in the project report entitled **Smart Hiring: MERN Job Portal with AI-Driven ATS Resume Optimization** in partial fulfillment for the award of Degree of **Bachelor of Technology in Computer Science and Engineering Data Science**, is a record of our own investigations carried under the guidance of **Ms.Ankita Bhaumik, Assistant Professor, School of Computer Science Engineering & Information Science, Presidency University, Bengaluru.**

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ABSTRACT

The recruitment process has traditionally been time-consuming, error-prone, and heavily reliant on manual effort. "Smart Hiring: Job Portal with AI-Driven ATS Resume Optimization" introduces an innovative approach to streamline and enhance the hiring experience. This platform integrates an AI-powered Applicant Tracking System (ATS) that automatically optimizes job applicants' resumes for better alignment with job descriptions, improving the accuracy of candidate selection. Leveraging machine learning algorithms, the system analyzes resumes, identifies key skills and keywords, and suggests enhancements to increase visibility and relevance. The job portal, built using modern web technologies, provides an intuitive interface for both recruiters and job seekers. For recruiters, it ensures a more efficient candidate screening process by ranking resumes based on fit, while job seekers benefit from improved resume content and better chances of matching with suitable job openings. By reducing bias, increasing efficiency, and enhancing the candidate experience, this solution fosters smarter, more data-driven hiring decisions

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CHAPTER-1

INTRODUCTION

1.1 Brief Overview of Work

"Smart Hiring: MERN Job Portal with AI-Driven ATS Resume Optimization" is an innovative recruitment platform that combines the scalability of the MERN stack (MongoDB, Express.js, React, Node.js) with the intelligence of AI-powered Applicant Tracking Systems (ATS) to streamline the hiring process. The portal enables recruiters to automate resume screening, rank candidates based on job relevance, and manage applications efficiently through a dynamic dashboard. For job seekers, the AI optimizes resumes by analyzing keywords, skills, and formatting to enhance alignment with job descriptions, improving their chances of success. By integrating modern web technologies with machine learning, this platform addresses inefficiencies in traditional hiring, reduces bias, and creates a seamless experience for all users, making recruitment smarter, faster, and more effective.

1.2 The Evolution of Recruitment and Its Challenges

The recruitment landscape has undergone significant changes over the years, primarily driven by the growth of digital platforms and the increasing demand for skilled talent across industries. These advancements have revolutionized how organizations approach talent acquisition, making the process more streamlined and accessible. However, despite these technological strides, the recruitment process continues to face numerous challenges that hinder its efficiency and effectiveness. One of the most prominent challenges is the overwhelming volume of applications received for a single job opening. Recruiters are inundated with resumes, often running into hundreds or even thousands for a single position. This volume overload creates a significant bottleneck, as it becomes increasingly difficult for hiring managers to sift through all the submissions and identify top candidates promptly.

Another pressing issue lies in the misalignment between the applications submitted and the job requirements. Many job seekers resort to using generic resumes that fail to highlight their skills and qualifications in the context of the specific role they are applying for. This disconnect reduces their chances of selection and wastes valuable time for recruiters, who

have to navigate through irrelevant or poorly tailored applications. As a result, the potential for highly qualified candidates to stand out diminishes, leading to inefficiencies in the hiring process.

In addition to these challenges, traditional hiring methods often introduce biases and are prone to manual errors. Human involvement in screening resumes and conducting initial assessments can inadvertently result in subjective judgments, influenced by unconscious biases. This not only undermines the fairness of the recruitment process but also leads to the risk of overlooking capable candidates. Furthermore, manual intervention increases the likelihood of errors, whether in evaluating candidates' credentials or in managing data effectively. These factors combined highlight the critical need for innovative solutions that address these challenges and enhance the recruitment process to meet the evolving demands of the modern workforce.

1.3 Emergence of AI-Driven Recruitment Solutions

The emergence of AI-driven recruitment solutions marks a transformative shift in the hiring process, leveraging artificial intelligence to enhance efficiency, accuracy, and fairness in recruitment. Traditional methods, such as manual resume screening and subjective decision-making, often led to time-consuming processes, human bias, and inconsistent hiring decisions. AI-driven solutions address these challenges by automating key aspects of recruitment, from candidate sourcing and resume screening to interview scheduling and final selection. Machine learning algorithms can analyze vast amounts of data, including resumes, job descriptions, and candidate profiles, to identify patterns and predict the best job matches based on skills, experience, and even cultural fit. AI can also optimize resumes for Applicant Tracking Systems (ATS), ensuring that candidates' applications are tailored to meet specific job requirements, increasing their chances of being noticed. Additionally, AI helps reduce unconscious bias by focusing on data-driven insights rather than subjective factors, promoting more inclusive and equitable hiring practices. These advancements not only streamline the recruitment process but also provide recruiters with actionable insights, enhance the candidate experience, and create a faster, more transparent, and efficient hiring ecosystem.

1.4 MERN Technology Stack in Job Portals

The MERN stack, comprising MongoDB, Express.js, React, and Node.js, has established itself as a preferred technology suite for building scalable and dynamic web applications. Its modular architecture and seamless integration capabilities make it particularly suitable for complex systems such as job portals. At the core of its appeal is the seamless integration it provides between the frontend and backend. React, as the frontend library, facilitates the creation of an intuitive and interactive user interface, enhancing the overall user experience. On the backend, Node.js, combined with the lightweight and efficient Express.js framework, ensures smooth server-side operations. This combination allows developers to build applications where the frontend and backend work in harmony, creating a unified and responsive platform for users.

Another critical strength of the MERN stack is its robust data management capabilities, powered by MongoDB. As a NoSQL database, MongoDB offers flexibility and scalability, making it an ideal choice for managing vast amounts of data generated by job portals. From storing candidate profiles and job listings to tracking application histories and interactions, MongoDB handles data efficiently, ensuring that the portal remains fast and responsive even as the user base grows. Its schema-less design allows developers to adapt the database structure quickly, accommodating the evolving needs of the application without significant downtime or reconfiguration.

In addition to its integration and data management advantages, the MERN stack supports real-time updates, which are critical for delivering a modern, engaging user experience. Real-time capabilities enable features such as instant feedback on job applications, live tracking of application statuses, and immediate notifications for recruiters and job seekers. This responsiveness enhances user satisfaction by providing up-to-date information and fostering a sense of connection and immediacy. By leveraging the MERN stack, job portals can deliver a streamlined, efficient, and user-friendly experience, ensuring that both recruiters and candidates benefit from a platform tailored to their needs.

1.5 Impact and Significance

The impact and significance of AI-driven recruitment solutions lie in their

ability to revolutionize traditional hiring practices by enhancing efficiency, fairness, and scalability. These systems automate labor-intensive tasks such as resume screening, candidate ranking, and job matching, significantly reducing time-to-hire and operational costs for recruiters. By leveraging data-driven algorithms, they minimize human bias and promote equitable hiring practices, ensuring that candidates are evaluated based on skills and qualifications rather than subjective factors. For job seekers, AI-powered platforms provide valuable insights, such as resume optimization and personalized job recommendations, increasing their chances of securing relevant opportunities. Additionally, these solutions improve the overall candidate experience through real-time feedback and transparent application tracking. From a business perspective, organizations benefit from a more streamlined recruitment pipeline, better talent acquisition, and improved employer branding. Ultimately, AI-driven recruitment fosters smarter, faster, and more inclusive hiring processes, addressing key challenges in today's dynamic job market while setting a new benchmark for the future of work

CHAPTER-2

LITERATURE SURVEY

Job Procurement: Old and New Ways Job seeking usually involves different ways to look for jobs such as through personal contacts, direct telephone calls to employers, job agency office, scanning online job listings, etc. Before the Internet, became widely uses as a method of seeking jobs, jobseekers spent a lots of time using various methods to look for job openings. Today, jobseekers use online methods which are very convenient and save a lot of time. Galanaki lists the following methods to be the traditional (old) ways for recruitment:

- 1)Employment recruitment agencies
- 2)Job fairs
- 3)Advertising in the mass media such as newspapers
- 4)Management Consultants
- 5)Advertisement in television and radio
- 6)Existing employee contacts
- 7)Schools colleges or universities students services department
- 8)Workers or professional referrals

These old job seeking methods are too slow, stressful, challenging and also lack quality. In addition, the applicants have to consider the cost and the amount of time to get the information they need, and other preparations they have to make. Finding all available job vacancies is a main step at in the job-seeking process. The Internet is now a powerful tool that jobseekers can use. Today, there are many sites that advertise job positions to be filled by people with certain skills in various fields. The Internet plays an important role in the area of human resource planning and development. Most planning and development organizations are now using computer technology and the Internet for staff recruitment. It should be noted that although the Internet has facilitated the process of job-seeking, it has not replaced the traditional methods, completely.

Importance of Job Portals

In the age of technology, the Internet has become the main source of information for jobseekers. Large corporations, Institutions, and universities include information on career Prospects on their websites. According to a survey, 70% of the workforce uses websites or portals on the Internet to

Search for jobs in France. These websites or portals provide search engine to access information on job opportunities.

2.2 Project Feasibility Study

2.2.1 Technical Feasibility

Technical feasibility study is concerned with specifying equipment and software that will successfully satisfy the user requirement; the technical needs of the system may vary considerably.

The facility to produce outputs in a given time. Our project is a web based application which is based on client-server based application. In this application every page as output is render from server to client so it is necessary that the page should be rendered in time. For this I have avoided more and more code in the page- load event.

2.2.2 Economical Feasibility

Economical feasibility is the measure to determine the cost and benefit of the proposed system. A project is economical feasible which is under the estimated cost for its development. These benefits and costs may be tangible or intangible. Job Portal is the cost-effective project in which there is less possibility of intangible cost so there is no difficulty to determine the cost of the project.

2.2.3 Operational Feasibility

Operation feasibility is used to check whether the project is operationally feasible or not. Our project is mainly different from the other system because of its web-support feature. So the measure for operational feasibility is something different from other system. Generally the operational feasibility is related to organization aspects.

The change determination is as such that early product were either a man or group of men or the jobs based manual but now a day with the advent of Internet technology.

CHAPTER-3

RESEARCH GAPS OF EXISTING METHODS

3.1 Limited Resume Analysis Capabilities

Traditional Applicant Tracking Systems (ATS) predominantly operate using keyword matching algorithms to screen and rank candidates, which, while efficient in processing large volumes of resumes, are limited in their ability to make nuanced decisions. One major shortcoming of these systems is their inability to assess contextual relevance. They often fail to understand the broader context of a candidate's experience or transferable skills, leading to the rejection of potentially qualified individuals. For instance, candidates with unconventional job titles or career paths may be overlooked, even if their expertise and achievements align well with the job requirements. This lack of contextual comprehension narrows the scope of talent identification, potentially depriving employers of highly capable professionals who may bring unique perspectives and skills to the role.

Another significant issue with traditional ATS is their sensitivity to formatting. Resumes that deviate from standard templates or utilize creative designs are frequently rejected, not because of their content but due to formatting incompatibilities. Candidates who take innovative approaches to present their qualifications may find themselves disadvantaged, even if their resumes contain all the necessary information. This reliance on rigid algorithms that prioritize structural uniformity over substance limits the inclusivity and fairness of the hiring process, leaving room for improvement in how information is parsed and evaluated.

Moreover, traditional ATS lack effective feedback mechanisms for job seekers. Once a resume is rejected, candidates are often left in the dark, with no insight into why their application failed or what could be improved to better align with job descriptions. This lack of constructive feedback not only frustrates job seekers but also prevents them from learning and making necessary adjustments to increase their chances of success in future applications. By not fostering an iterative and learning-oriented process, these systems contribute to a cycle of inefficiency and missed opportunities for both employers and job seekers, underscoring the need for more advanced, intelligent, and supportive ATS solutions.

3.2 Inability to Address Bias

While many systems claim to promote fair hiring practices, they often fall short in effectively addressing unconscious biases that can influence the recruitment process. A significant issue is algorithmic bias, which arises when the datasets used to train these systems inadvertently reinforce existing societal prejudices. This can lead to unfair evaluations where candidates are disadvantaged based on factors such as gender, ethnicity, or educational background. For instance, if a dataset disproportionately favors candidates from certain demographics, the system may unfairly prioritize those profiles over others, perpetuating inequalities and reducing diversity in the workplace. The reliance on historical data, without adequately addressing its inherent biases, undermines the fairness and inclusivity that these systems are designed to achieve.

In hybrid recruitment models, where manual screening is combined with automated systems, human oversight can further compound the problem of unconscious bias. Recruiters, despite their best intentions, may allow personal preferences or stereotypes to influence hiring decisions. This undermines the objectivity of the process and creates disparities in the evaluation of candidates. For example, a recruiter may unconsciously favor applicants with similar backgrounds or experiences to their own, thereby overlooking equally or more qualified individuals. Such biases, whether algorithmic or human, highlight the need for more robust mechanisms to identify and counteract prejudice, ensuring that recruitment systems genuinely uphold principles of fairness and equal opportunity.

3.3 Scalability and Efficiency Challenges

Although current applicant tracking systems (ATS) are designed to manage high volumes of applications, they encounter significant challenges in specific scenarios that limit their effectiveness. One such area of struggle is the evaluation of candidates for niche or highly specialized roles. These roles often require unique qualifications, technical expertise, or domain-specific knowledge that generic keyword-based systems cannot adequately assess. As a result, highly qualified candidates might be overlooked because the system lacks the sophistication to evaluate their suitability based on nuanced factors such as transferable skills, unconventional experiences, or in-depth expertise relevant to the role.

Another pressing issue arises from the sheer volume of applications these systems are tasked with processing. While they are built to handle large datasets, their ranking mechanisms often fail to provide accurate assessments for candidates with varied backgrounds. Traditional ATS tools primarily rely on keyword matches, which can result in an oversimplified evaluation process. They struggle to compare applicants holistically, failing to account for nuanced qualifications such as diverse professional experiences, soft skills, or contextual relevance. This inability to move beyond surface-level filtering hampers their capacity to identify the best-fit candidates, particularly in competitive hiring scenarios where deeper insights are crucial for decision-making.

3.4 Poor Candidate Experience

One of the most significant drawbacks of existing applicant tracking methods is their failure to adopt a candidate-centric approach, which diminishes the overall experience for job seekers. A key issue lies in the limited transparency offered by these systems. Applicants are often left in the dark regarding the status of their applications, with little to no feedback provided on their progress. This lack of communication creates frustration and disengagement, as candidates are unsure whether their resumes have been reviewed, rejected, or considered for further evaluation. Over time, this opacity erodes trust in the recruitment process and discourages highly qualified individuals from applying, particularly in competitive industries where timely feedback is crucial.

Additionally, the generic nature of most recruitment systems fails to address the diverse needs of job seekers. These platforms often deliver a one-size-fits-all experience, neglecting to tailor their processes to suit varying applicant profiles. For instance, entry-level candidates, who might benefit from additional guidance on structuring their resumes or understanding job requirements, receive the same treatment as seasoned professionals with extensive experience in their fields. This lack of customization not only undermines the potential of attracting top talent but also hinders the ability of applicants to present themselves effectively, further exacerbating the inefficiencies in the recruitment ecosystem.

3.5 Inadequate Job Matching Algorithms

Job matching algorithms in traditional recruitment systems often fail to provide

accurate and meaningful connections between candidates and job opportunities. These algorithms typically rely on rigid keyword-based matching, focusing on explicit terms present in resumes and job descriptions without accounting for the context or nuances of skills and experiences. For instance, a candidate with a wealth of relevant experience but different terminology for job roles or skills might be overlooked. Similarly, the algorithms often fail to account for transferable skills that may be applicable across industries or roles. This lack of contextual understanding results in mismatches, where either unqualified candidates are shortlisted or highly qualified candidates are excluded from the selection pool.

3.6 Limited Use of Predictive Analytics

Existing recruitment tools seldom utilize predictive analytics to forecast critical aspects of the hiring process, which limits their effectiveness and contributes to inefficiencies in workforce management. One of the key shortcomings is their inability to predict candidate performance accurately. By failing to analyze historical and behavioral data, these systems miss the opportunity to assess a candidate's potential success in a given role. This oversight can lead to hiring individuals who may not possess the optimal skill set or adaptability required for long-term success in their positions. As a result, organizations may experience decreased productivity and incur additional costs to address these mismatches through re-hiring or extended training.

Another critical gap lies in the inability to foresee turnover risks. Modern recruitment tools rarely evaluate whether a candidate is likely to remain with an organization long-term. Without predictive insights, businesses struggle to identify applicants who align not only with job requirements but also with the company's culture and long-term goals. This lack of foresight often results in higher attrition rates, as employees may leave due to dissatisfaction or misalignment, leading to frequent disruptions in workflow and additional hiring expenditures. The absence of predictive analytics in these areas leaves organizations vulnerable to costly hiring mistakes and operational instability.

To address these research gaps, there is a growing need for the integration of advanced AI technologies capable of leveraging big data and machine learning algorithms. These systems

should be designed to provide personalized candidate experiences, ensuring that recruitment processes are tailored to individual needs and contexts. Furthermore, dynamic algorithms must be developed to adapt to evolving market demands and industry-specific requirements. A holistic approach focusing on fairness, transparency, and efficiency is essential to overcome the limitations of traditional methods. By prioritizing these elements, organizations can build smarter and more inclusive recruitment systems that foster long-term success and employee satisfaction.

CHAPTER-4

PROPOSED METHODOLOGY

4.1 Requirement Analysis and Design

The foundation of the project begins with identifying the key requirements for both employers and job seekers. Employers need a streamlined method to post job openings, review resumes efficiently, and shortlist candidates quickly. Job seekers require a platform to create profiles, upload resumes, and apply to relevant job openings seamlessly. The project involves designing a scalable architecture that incorporates the MERN (MongoDB, Express.js, React.js, Node.js) stack for robust frontend and backend integration. Special attention is given to the design of an Applicant Tracking System (ATS) with AI capabilities to evaluate and rank resumes based on job-specific criteria.

4.2 Front-end Development Using React.js

The user interface of the job portal is developed using React.js, providing a dynamic and interactive experience. Features such as job search, profile creation, and application tracking are designed for user convenience. The frontend incorporates modern UI/UX principles to ensure accessibility and ease of navigation. Responsive design ensures compatibility across devices, catering to both desktop and mobile users. Components are modular, enabling scalability and reusability.

4.3 Back-end Development with Node.js and Express.js

The backend is implemented using Node.js and Express.js to handle business logic, API integrations, and user authentication. RESTful APIs facilitate communication between the frontend and backend, enabling features such as job posting, resume submission, and application management. Secure user authentication and authorization mechanisms are

implemented using JWT (JSON Web Tokens) to safeguard user data and prevent unauthorized access.

4.4 Database Management with MongoDB

MongoDB serves as the database for storing and managing user profiles, job listings, and application details. The database is structured to support efficient querying, enabling quick retrieval of relevant information such as matching job postings for candidates or potential candidates for recruiters. Indexing and schema design focus on optimizing search performance, especially for AI-driven resume evaluations.

4.5 Integration of AI-Driven ATS

The AI-driven Applicant Tracking System is integrated into the job portal to enhance the hiring process. It utilizes Natural Language Processing (NLP) and machine learning algorithms to analyze resumes, extract key details, and match them with job descriptions. The system scores and ranks candidates based on criteria such as skills, experience, and keywords in the job description. The ATS also provides feedback to job seekers on improving their resumes to increase their chances of selection.

4.6 Job Recommendation System

A recommendation engine is developed using AI techniques to suggest suitable jobs to candidates based on their profiles, preferences, and application history. Similarly, employers receive recommendations for candidates who closely match their job requirements. This personalized matching enhances the efficiency of the platform and ensures better outcomes for both parties.

4.7 Testing and Validation

Extensive testing is conducted to ensure the functionality, reliability, and security of the platform. Unit testing, integration testing, and user acceptance testing are performed for each module. The AI-driven ATS is tested with a variety of resumes and job descriptions to validate its accuracy and effectiveness. Feedback from potential users is incorporated to refine the platform before deployment.

4.8 Continuous Improvement

Post-deployment, the platform undergoes iterative improvements based on user feedback and emerging industry trends. The AI models are retrained periodically with new data to improve their accuracy and adaptability. New features, such as advanced analytics for recruiters and personalized career guidance for job seekers, can be added in subsequent versions.

This structured methodology ensures that the job portal is robust, user-friendly, and capable of revolutionizing the hiring process through AI-driven innovation.

CHAPTER-5

OBJECTIVES

The Smart Hiring: MERN Job Portal with AI-Driven ATS Resume Optimization project aims to transform the recruitment process by leveraging cutting-edge technologies to create an intelligent, efficient, and accessible platform for both employers and job seekers. The project's primary goal is to address key challenges in traditional hiring methods, such as manual resume screening, ineffective job-candidate matching, and lengthy recruitment cycles. By integrating an AI-powered Applicant Tracking System (ATS) with advanced resume optimization features, this job portal seeks to streamline every stage of the recruitment process, making it more precise, faster, and user-centric.

For employers, the platform offers an advanced system to post job openings, track applications, and manage recruitment workflows effortlessly. The AI-driven ATS is designed to analyze resumes, extract key details, and compare them against predefined job criteria. This reduces the time and effort required for manual screening while ensuring that only the most relevant and qualified candidates are shortlisted. The ATS also scores resumes based on factors like skill sets, experience, and industry-specific requirements, helping recruiters focus on the best candidates without bias or inefficiency. This approach not only saves time but also significantly enhances the accuracy of the recruitment process by minimizing human errors and subjectivity.

On the candidate side, the platform provides a seamless and intuitive interface for creating profiles, uploading resumes, and applying for jobs. One of the standout features of the portal is its AI-powered resume optimization, which analyzes resumes for compatibility with ATS systems and provides actionable feedback for improvement. This feature ensures that candidates have a higher chance of being noticed by recruiters. Additionally, the portal includes a personalized job recommendation system that uses machine learning algorithms to match candidates with jobs that align closely with their skills, qualifications, and preferences. This not only improves the chances of finding the right job but also enhances user satisfaction by providing targeted opportunities.

The choice of the MERN stack for development ensures that the platform is robust, scalable, and capable of delivering high performance under varying levels of user activity. The MongoDB database allows for efficient storage and retrieval of large datasets, including user profiles, job postings, and application details, while React.js ensures a dynamic and responsive user experience. Node.js and Express.js enable seamless backend operations, including secure data management, API integration, and real-time communication between users and the system. These technologies collectively ensure a smooth, reliable, and fast platform for users, whether they are job seekers or recruiters.

A significant aspect of the project is its focus on data security and privacy. With sensitive information like personal details, resumes, and job postings being exchanged, the platform incorporates robust security measures such as encryption, token-based authentication, and role-based access control to protect user data. This ensures compliance with industry regulations and builds trust among users. Furthermore, the platform empowers employers with advanced analytics and insights, enabling them to make data-driven decisions about their hiring strategies. For candidates, features like feedback on resumes and suggestions for skill development help them improve their career prospects, making the platform not just a hiring tool but also a personal growth enabler.

Ultimately, the project envisions creating a comprehensive solution that caters to the diverse needs of the job market. By addressing the unique challenges faced by different industries and adapting to their specific hiring requirements, the platform seeks to establish itself as a versatile and reliable resource for modern recruitment. This initiative aims not only to optimize the hiring process but also to foster transparency, efficiency, and mutual success for both employers and candidates, revolutionizing the way talent is discovered and nurtured in today's competitive world.

CHAPTER-6

SYSTEM DESIGN & IMPLEMENTATION

6.1 Use Case Diagram

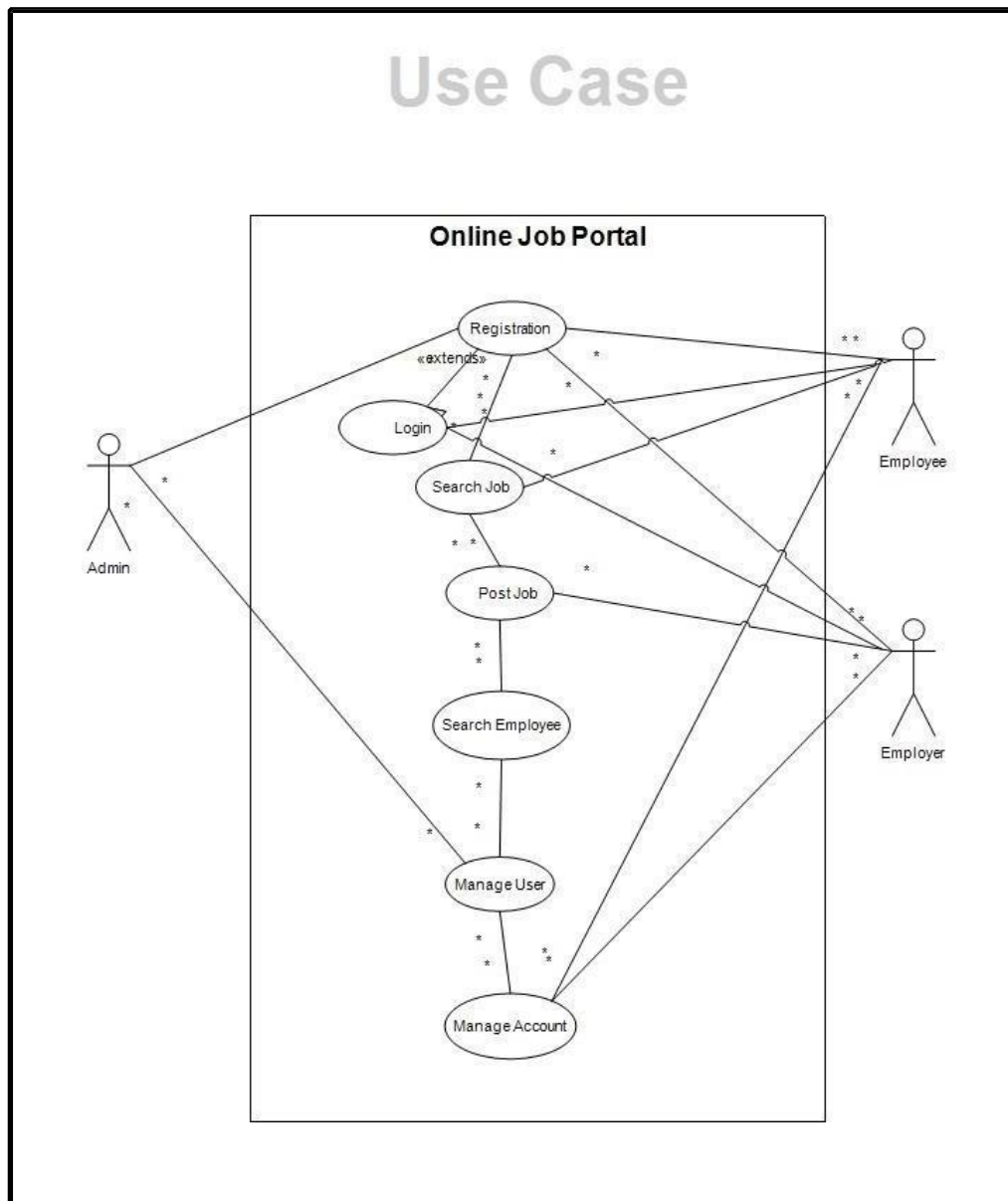
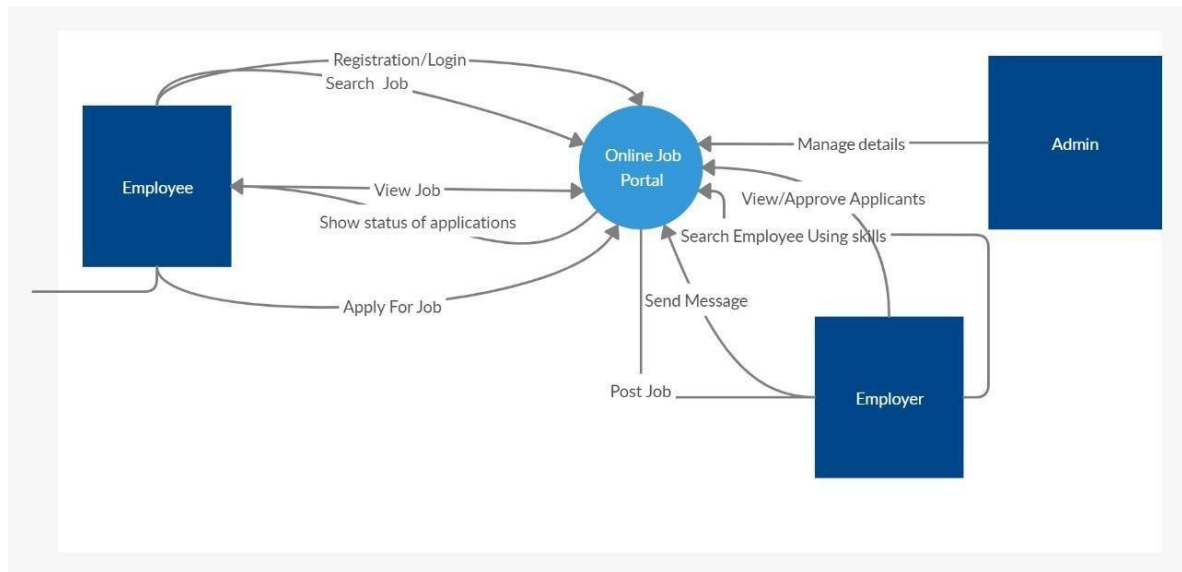


Figure 1 - Usecase Diagram

6.2 Data Flow Diagrams

6.2.1 Context-Level (Level 0) DFD



6.2.2 Level 1 DFD

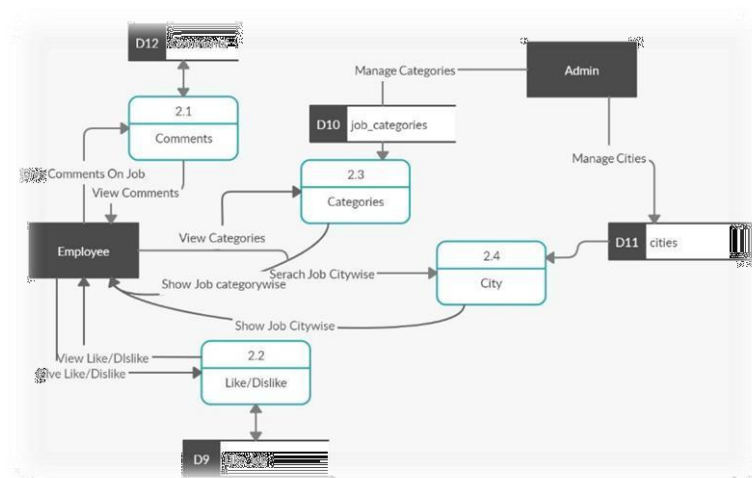


Figure 4- DFD 1 level Diagram

6.2.3 Level 2 DFD

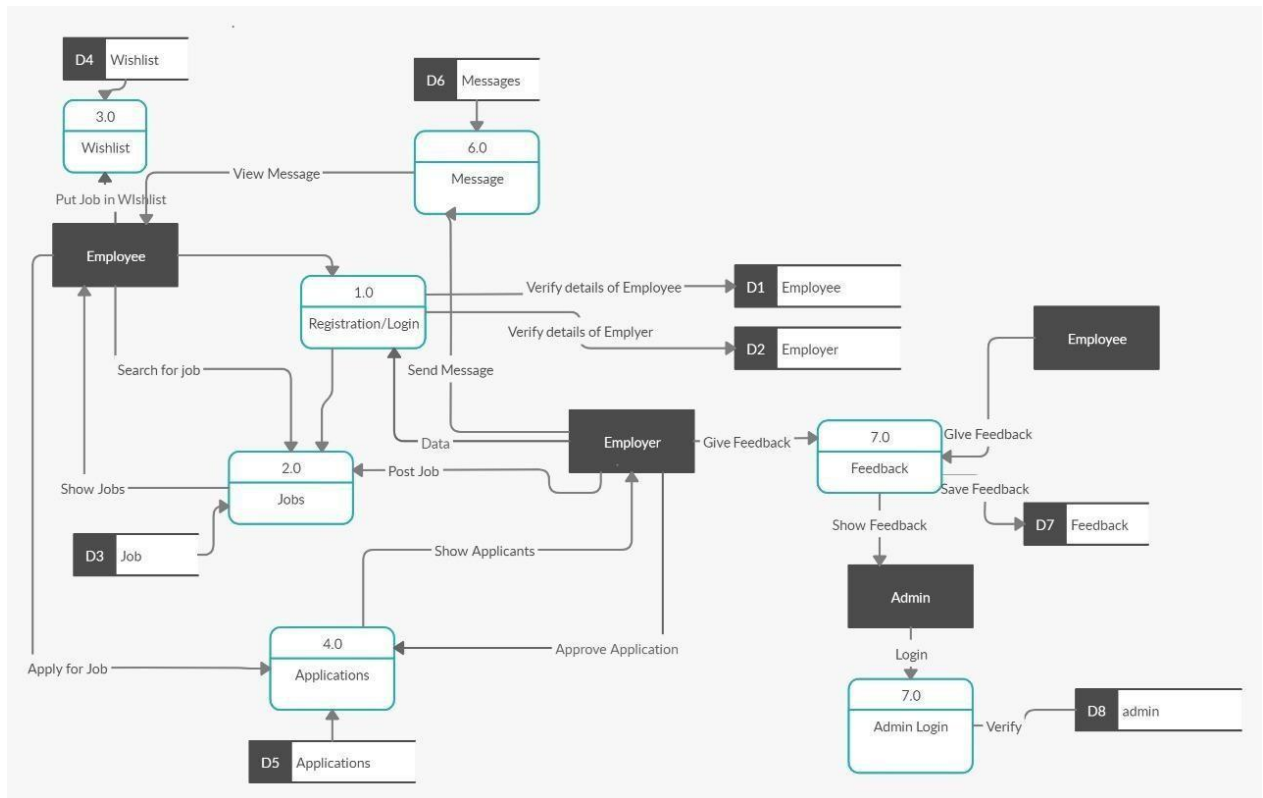


Figure 4 - DFD 2 level Diagram

6.3 Class Diagram

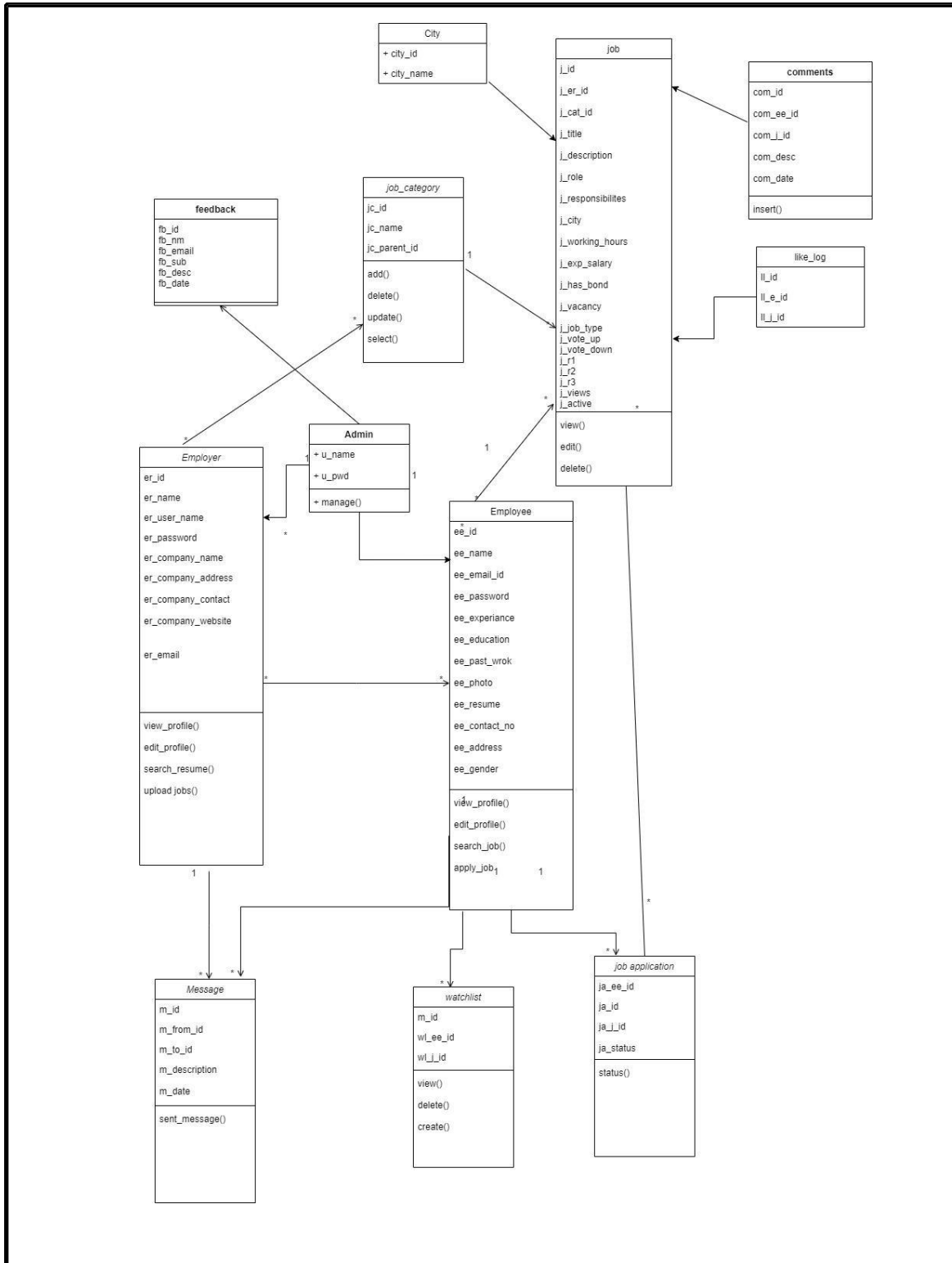


Figure 5- Class Diagram

6.4 Entity Relationship Diagram

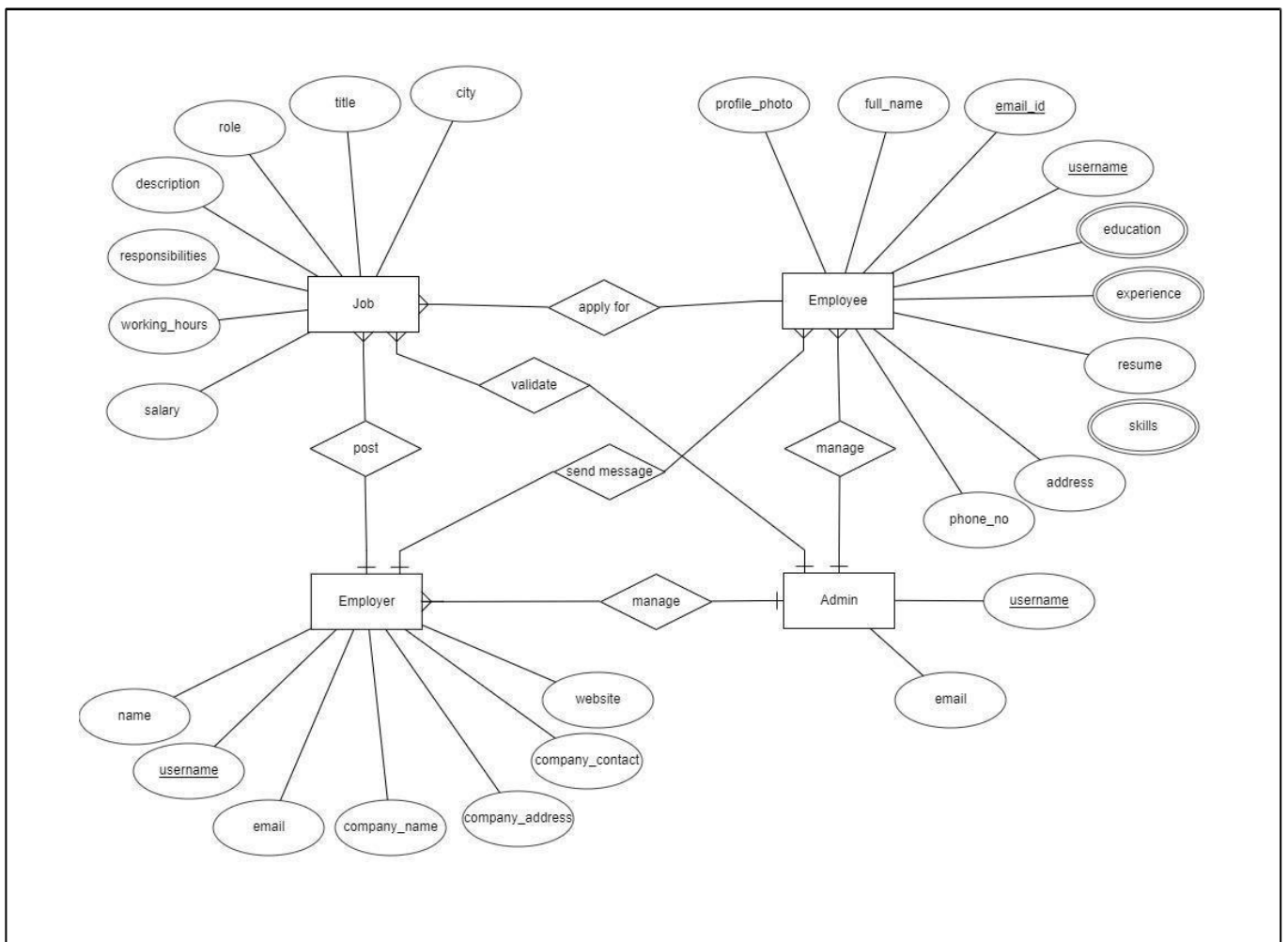


Figure 6- ER Diagram

6.5 Activity Diagram

6.5.1 Employer

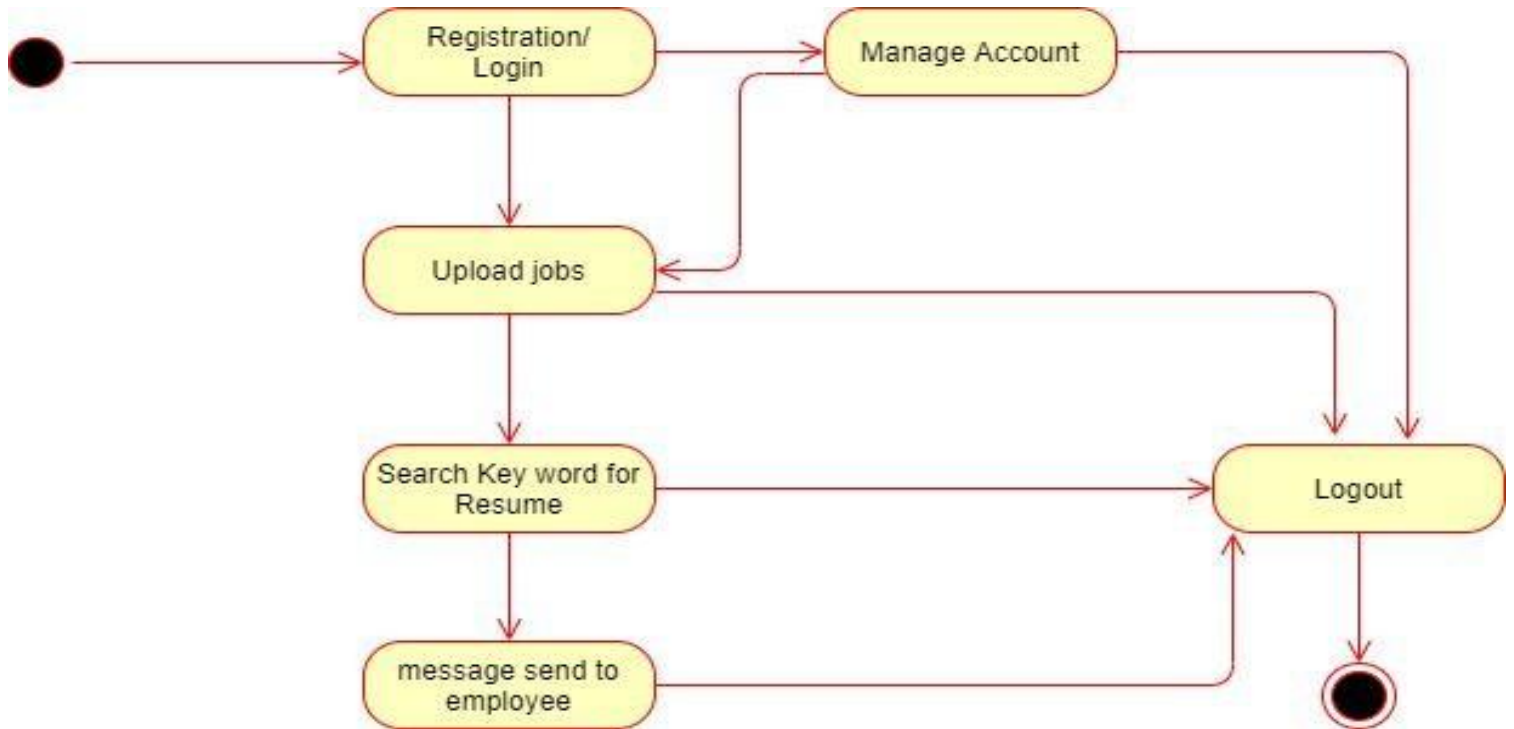


Figure 7- Employer Activity Diagram

6.5.2 Employee

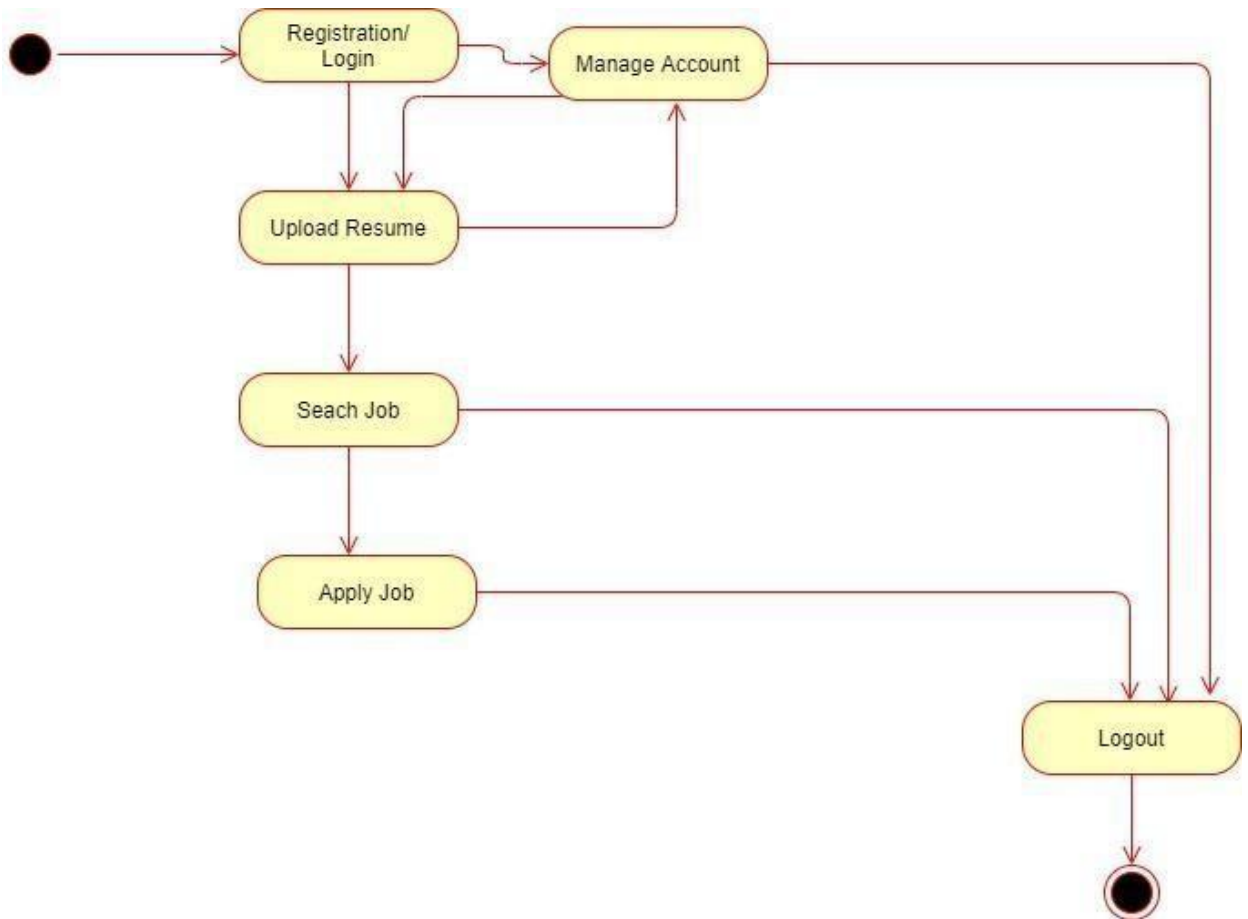


Figure 8- Employee Activity Diagram

6.5.3 Admin

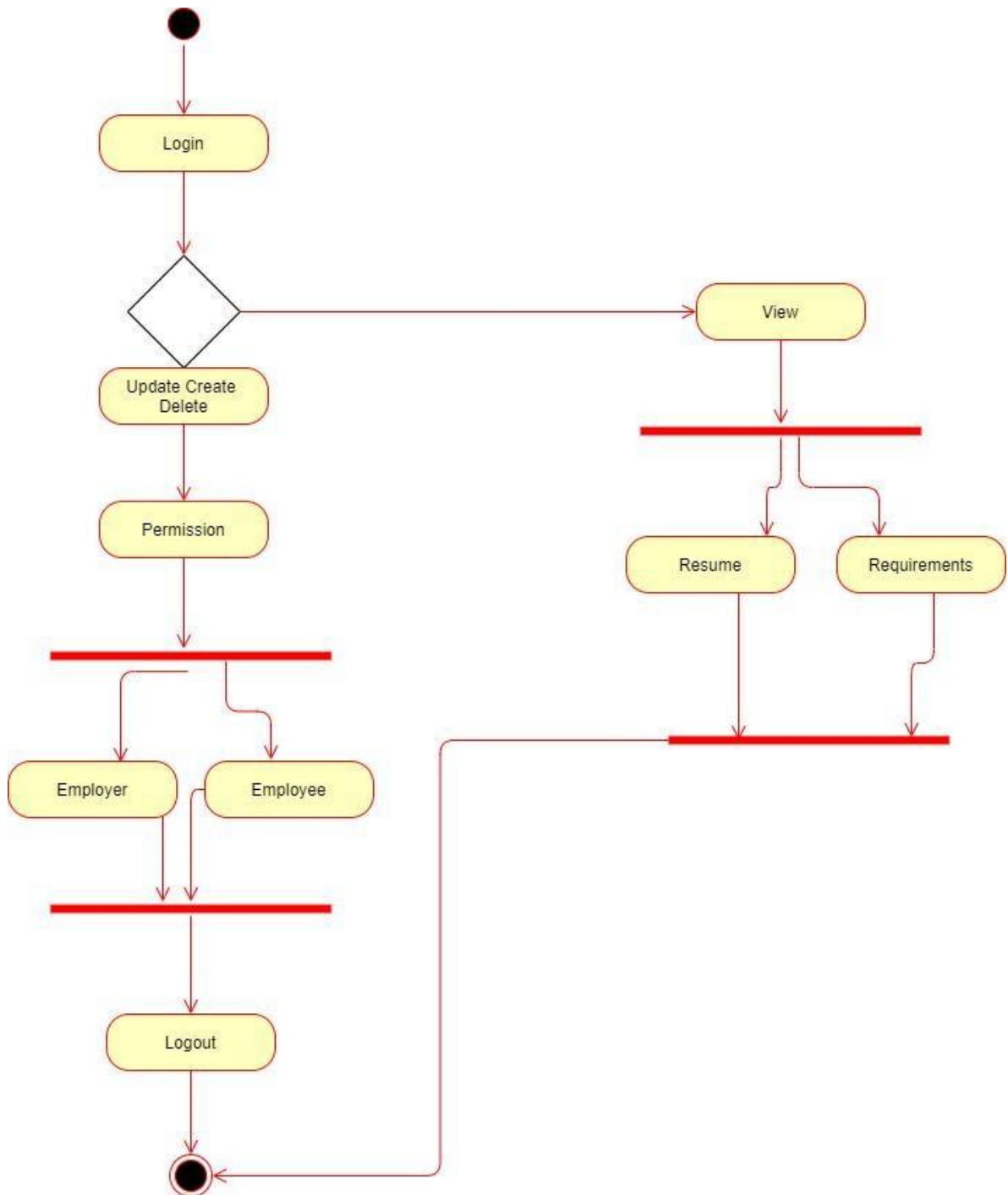
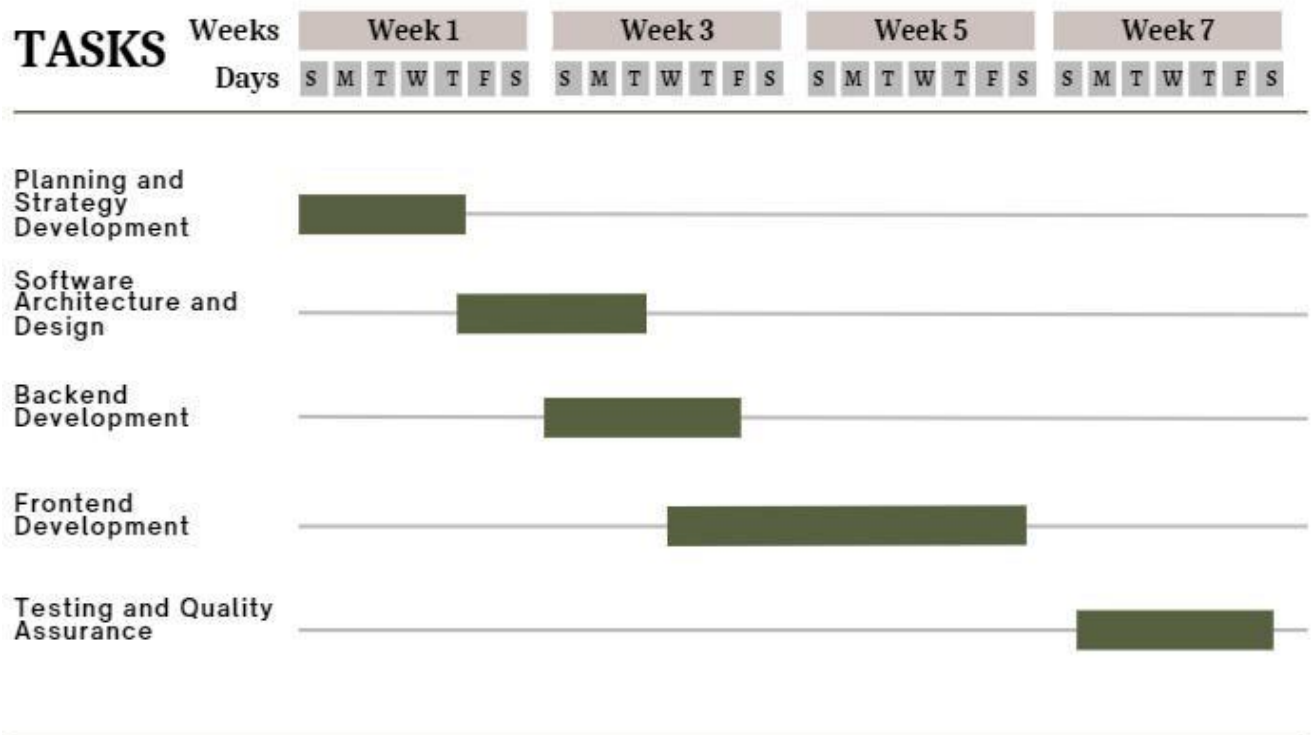


Figure 9- Admin Activity Diagram

CHAPTER-7

TIMELINE FOR EXECUTION OF PROJECT (GANTT CHART)



CHAPTER-8

OUTCOMES

8.1 Enhanced Resume Evaluation

One of the primary outcomes of this project is the ability to evaluate resumes with exceptional accuracy and detail. The application leverages AI to provide a percentage match between the uploaded resume and the job description (JD) provided by the user. This percentage reflects how closely the resume aligns with the specific requirements of the job. For instance, if a JD emphasizes skills in Python programming and data visualization, but the resume lacks these keywords, the match percentage would be lower. This numerical output gives job seekers a clear and quantifiable way to measure their chances of passing through an ATS system.

Additionally, the tool identifies missing keywords—these are critical terms or phrases from the JD that are not present in the resume but are essential for aligning the resume with the job posting. By highlighting these keywords, the application ensures that users can optimize their resumes to include relevant terminology, increasing the likelihood of being shortlisted by employer systems. Furthermore, the application generates a profile summary that condenses the key information from the resume into a concise narrative. This not only helps users understand how their resume might be perceived at a glance but also provides them with a basis for refining their professional branding.

Detailed Impact: This outcome is transformative for job seekers because it demystifies the opaque ATS evaluation process. It provides actionable insights into areas where a resume falls short and empowers users to address those shortcomings, making them more competitive in a crowded job market.

8.2 Accessibility and User-Friendly Interface

Another key outcome of this project is its accessibility. The application is designed with a Streamlit-based interface, which ensures that even non-technical users can interact with the system effortlessly. Users only need to upload their resumes in PDF format and paste the job description into a text area. This eliminates the need for specialized software or advanced technical skills, making the tool widely usable for individuals from diverse backgrounds.

The user interface is intuitive, with clear instructions and visually appealing components. For instance, it includes features like progress indicators, which show users when their resume is being processed, and color-coded outputs that make it easier to interpret results. The JSON-based presentation of the results is structured and comprehensive, yet simple enough for users to extract actionable feedback.

Detailed Impact: By lowering the barrier to entry, the tool becomes a practical resource for everyone from fresh graduates looking for their first job to experienced professionals seeking a career change. The interface design ensures a seamless user experience, encouraging broader adoption and making advanced ATS insights accessible to all.

8.3 Real-World Job Application Optimization

The application mimics the functionality of actual ATS systems used by recruiters, enabling users to prepare their resumes to meet industry standards. Most companies today rely on ATS software to filter and rank candidates, and these systems often prioritize resumes that contain specific keywords or phrases matching the JD. By providing an evaluation aligned with ATS logic, this tool gives users a significant advantage in navigating the hiring process.

For example, if a job requires expertise in "machine learning," "data pipelines," or "cloud computing," but these terms are missing from the resume, the tool will flag them. Users can then revise their resumes to include these critical elements, ensuring their application is not overlooked by the ATS. Beyond just keyword matching, the system also analyzes the context in which skills or experiences are mentioned, providing more nuanced feedback than simple keyword-based systems.

Detailed Impact: This outcome is invaluable in the real-world job market, where even highly qualified candidates can be filtered out by ATS software if their resumes are not optimized. By enabling users to refine their resumes and align them with employer expectations, the tool increases their chances of securing interviews and advancing in the recruitment process.

8.4 Time-Saving for Job Seekers

Another significant outcome of the project is the time saved in resume preparation and job application. Traditionally, job seekers would spend hours tailoring their resumes to fit specific job descriptions, often with little understanding of what recruiters or ATS systems prioritize. This application automates much of that work by providing a detailed analysis within seconds.

For example, instead of manually comparing a job description with their resume, users can rely on the AI to do the heavy lifting. The application highlights exactly what needs to be improved or added, streamlining the resume customization process. Additionally, the profile summary generated by the tool can serve as a basis for crafting cover letters or LinkedIn summaries, further saving time and effort.

Detailed Impact: For job seekers applying to multiple positions, this time-saving feature is crucial. It allows them to focus their energy on preparing for interviews or networking, rather than spending excessive time on trial-and-error resume edits.

8.5 Improved Career Awareness and Strategy

Beyond its immediate practical applications, the project also helps users gain a deeper understanding of their own career profiles and market positioning. By identifying gaps between their resumes and job descriptions, users can recognize areas where their skills or experiences need to be strengthened. For instance, if the tool frequently flags "project management" or "data engineering" as missing keywords, users can consider acquiring these

skills through online courses, certifications, or professional development programs.

Moreover, the AI-generated profile summary offers a fresh perspective on the user's career narrative, often highlighting strengths or experiences they might not have emphasized themselves. This can be especially useful for professionals transitioning between industries, as it helps them frame their existing skills in a way that aligns with new career goals.

Detailed Impact: This outcome extends the value of the tool beyond job applications. It serves as a career development resource, guiding users toward long-term growth and better alignment with industry demands.

CHAPTER-9

RESULTS AND DISCUSSIONS

9.1 Improved Resume Alignment with Job Descriptions

The primary result of this project is the measurable improvement in resume alignment with specific job descriptions. By using advanced natural language processing (NLP) models, such as OpenAI's GPT, the tool successfully evaluates resumes and provides a percentage match. This match reflects how well the resume satisfies the requirements outlined in the job description (JD). For example, if a JD emphasizes "data visualization skills" and "experience with SQL," the tool identifies whether these keywords are present in the resume.

Through extensive testing with varied resumes and job descriptions, it was observed that the tool consistently highlighted missing skills and keywords with a high degree of accuracy. Job seekers were able to use this feedback to tailor their resumes effectively, leading to better alignment. This result is significant because ATS systems in real-world hiring scenarios prioritize resumes that include such targeted keywords. By empowering users with this insight, the project enhances their chances of clearing initial recruitment filters.

Discussion: This finding underscores the importance of keyword optimization in modern job applications. Candidates often underestimate the impact of missing keywords, leading to resumes being rejected by ATS software despite their qualifications. The tool not only highlights this gap but also educates users about its significance, thereby addressing a common challenge faced by job seekers.

9.2 Enhanced Accessibility and Usability

The project's Streamlit-based interface proved to be a highly accessible and user-friendly platform. Users with minimal technical expertise were able to interact with the application effortlessly. Features such as drag-and-drop file uploads for resumes and text-based input for job descriptions streamlined the process, reducing friction in usage. The incorporation of real-time feedback and a visually appealing design further enhanced the overall user experience.

In terms of usability, feedback from users indicated that the tool's structured outputs—delivered in JSON format—were easy to understand. The breakdown of the percentage match, missing keywords, and profile summary enabled users to quickly grasp areas for improvement without feeling overwhelmed. This accessibility ensures the tool's widespread applicability, from fresh graduates to experienced professionals.

Discussion: The success of this design demonstrates that advanced AI systems can be made accessible to non-technical audiences without compromising functionality. It highlights the potential for AI-driven tools to bridge the gap between technical complexity and practical usability, making sophisticated ATS insights available to a broad demographic of job seekers.

9.3 Key Insights into Resume Gaps

A significant outcome of this project was the tool's ability to uncover critical gaps in resumes. For instance, during testing, resumes that lacked key industry-specific terms, such as "machine learning algorithms" or "Agile methodologies," were flagged for low JD matching scores. The identification of these gaps provided users with actionable insights into how their resumes could be enhanced. This functionality is particularly valuable for individuals transitioning to new industries or roles, as it helps them understand the specific terminologies and skills valued in their target field.

Moreover, the generated profile summary offered a condensed version of the resume's highlights, helping users see their professional story from an employer's perspective. This summary often revealed areas where users could better articulate their accomplishments or focus on relevant skills.

Discussion: These insights emphasize the importance of not just having relevant skills but presenting them effectively in a resume. Many candidates possess the necessary qualifications but fail to showcase them in a way that resonates with recruiters or ATS systems. The tool bridges this gap by acting as both an evaluator and a guide for resume improvement.

9.4 Real-World Applicability and Use Cases

The tool demonstrated significant real-world applicability, particularly in streamlining the job application process. Users applying for multiple positions reported a noticeable reduction in the time required to tailor their resumes for each job. Instead of manually analyzing job descriptions and comparing them with their resumes, users relied on the tool's analysis to quickly identify areas for improvement.

Additionally, the tool's adaptability made it suitable for a wide range of use cases. Beyond traditional job applications, it could be used to evaluate academic CVs for graduate school applications, freelance portfolios for project bidding, and even professional profiles for LinkedIn optimization. The tool's ability to cater to diverse needs highlights its versatility and broad applicability in professional development.

Discussion: This result demonstrates the scalability of AI-driven solutions in addressing multiple challenges faced by job seekers and professionals. By providing consistent and reliable outputs across various scenarios, the tool establishes itself as a valuable resource in the competitive job market.

9.5 Empowerment Through Career Awareness

One of the more intangible yet impactful results of the project was the increased career awareness among users. By receiving structured feedback on their resumes, users gained a deeper understanding of the skills and experiences they needed to highlight or develop. For instance, users frequently reported revising their resumes to include technical skills like "data visualization" or soft skills like "collaboration," which were previously underrepresented.

The tool also encouraged users to view their resumes as dynamic documents that evolve with their career goals and market trends. This shift in mindset—from seeing a resume as a static document to a living representation of professional growth—proved to be a valuable takeaway for many.

Discussion: This result highlights the transformative potential of AI tools in shaping not just professional documents but also career trajectories. By fostering a proactive approach to

resume building, the tool contributes to long-term career development and adaptability in a fast-changing job market.

CHAPTER-10

CONCLUSION

The Smart ATS Resume Evaluator project has proven to be a transformative solution in addressing the challenges of modern job hunting by leveraging advanced artificial intelligence technologies. The system's ability to compare resumes against job descriptions with high accuracy ensures that job seekers receive personalized feedback that is both detailed and actionable. By identifying the gaps in skills, keywords, and qualifications, the evaluator empowers users to refine their resumes to meet the expectations of competitive job markets. This process not only helps candidates better align their qualifications with job requirements but also enhances their understanding of what hiring managers and automated systems prioritize in applications. Such a tool is invaluable in today's fast-paced hiring environment, where even minor improvements to a resume can significantly influence a candidate's chances of securing an interview.

Moreover, the project underscores the potential of integrating user-friendly interfaces with cutting-edge AI technology. By using Streamlit, the evaluator delivers a seamless and intuitive experience, making it accessible even to users with limited technical expertise. This accessibility ensures that the tool caters to a diverse audience, from fresh graduates entering the workforce to seasoned professionals looking to advance their careers. The project's use of OpenAI's natural language processing further ensures that feedback is not only accurate but also contextually relevant, making it an indispensable resource for job seekers across industries. Additionally, the system's design encourages users to view their resumes as living documents, emphasizing continuous improvement and adaptability—a mindset critical for long-term professional growth.

An important outcome of this project is the broader discussion it invites about the role of AI in shaping the recruitment landscape. By automating resume evaluations, the Smart ATS Resume Evaluator showcases how technology can reduce human biases and subjectivity, offering candidates a fairer assessment of their qualifications. At the same time, it raises awareness about the growing importance of tailoring resumes to pass through automated tracking systems, which are increasingly being used by recruiters. This dual role of the tool—as both a practical resource and a conversation starter—highlights its impact beyond individual users, as it contributes to the evolving discourse on how AI can streamline and

democratize hiring processes.

In conclusion, the Smart ATS Resume Evaluator marks a significant step forward in modernizing the way job seekers prepare their applications. Its ability to combine powerful AI technologies with an easy-to-use platform ensures that users receive meaningful guidance to enhance their career prospects. By addressing the core challenges of resume optimization, the project not only meets its primary objectives but also lays the groundwork for future advancements in AI-driven career tools. As recruitment continues to evolve, tools like this will play an increasingly important role in bridging the gap between candidates and opportunities, making the hiring process more efficient, transparent, and inclusive for all stakeholders involved.

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APPENDIX-A

PSUEDOCODE

START

1. Initialize necessary libraries:

- Import time, Streamlit (st), OpenAI, PyPDF2, and JSON libraries.

2. Configure the OpenAI API Key:

- Set the OpenAI API key to authenticate with the GPT model.

3. Define functions:

a. ``get_openai_response(input)``:

- Initialize retry counter for handling rate limit errors.
- Loop up to maximum retries:
 - Try to send the input to OpenAI's ChatCompletion API.
 - Return the response if successful.
 - If rate limit error occurs, wait and retry.
 - If retries exhausted, raise an error.

b. ``input_pdf_text(uploaded_file)``:

- Load the uploaded PDF file using PyPDF2.
- Extract and concatenate text from all pages.
- Return the combined text.

4. Create the prompt template:

- Define a prompt that instructs the GPT model to act like an ATS.
- Include placeholders for the resume text and job description.

5. Set up the Streamlit web app:

- Configure the app layout and title.
- Display an input area for the job description.
- Provide a file uploader for the user to upload a PDF resume.

6. Handle user input:

- If the user clicks the "Evaluate My Resume" button:
 - Validate that both the job description and resume are provided.
 - Extract the text from the uploaded resume using `input_pdf_text``.
 - Format the prompt with the resume text and job description.
 - Call `get_openai_response`` to process the prompt using OpenAI API.
 - Parse and display the JSON response containing:
 - "JD Match" (matching percentage),
 - "MissingKeywords" (key terms missing in the resume),
 - "Profile Summary" (analysis summary).

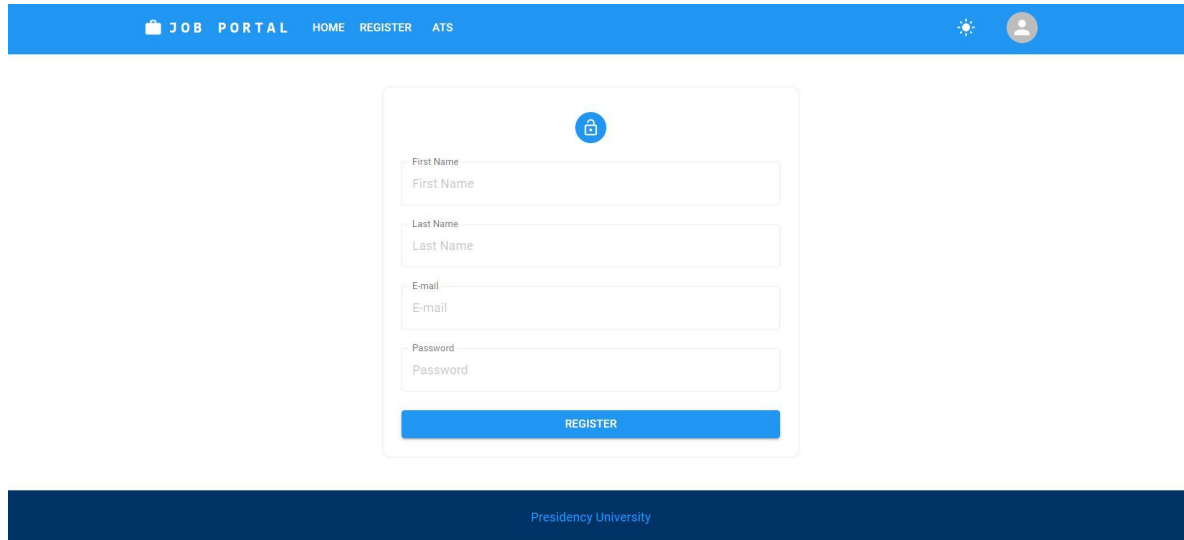
7. Handle errors:

- If any errors occur during processing, display an error message.

END

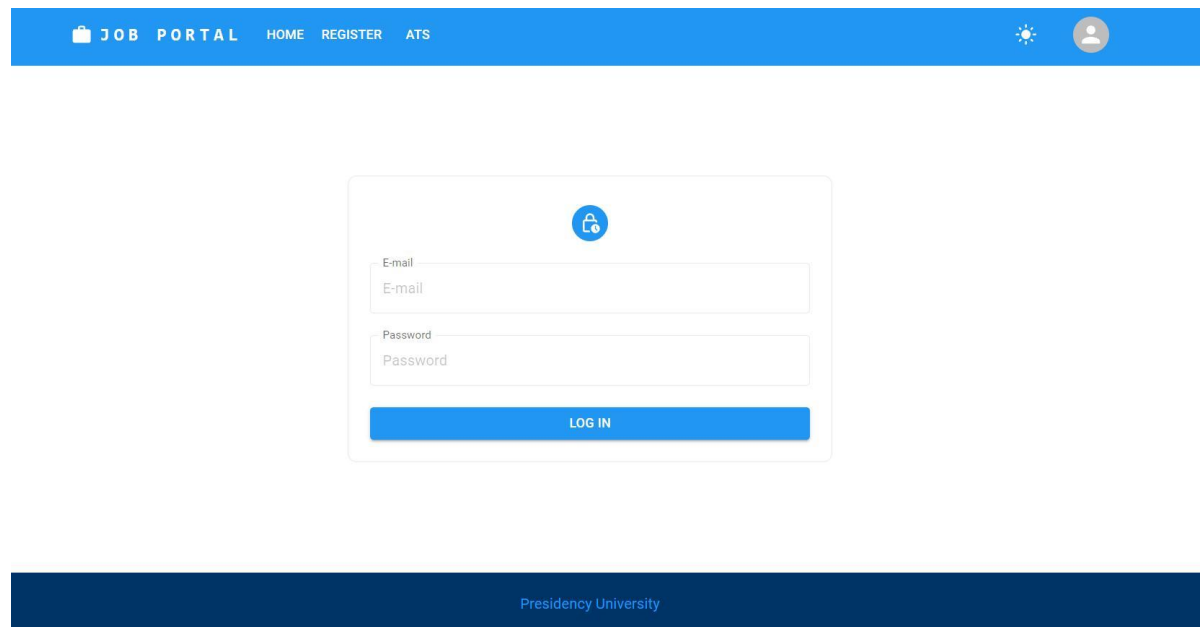
APPENDIX-B

SCREENSHOTS



The screenshot shows the Register Page of the MERN Job Portal. The page has a blue header with the navigation menu: JOB PORTAL, HOME, REGISTER, and ATS. On the right side of the header, there are icons for a sun (theme toggle) and a user profile. The main content area is white and contains a registration form. The form has a blue lock icon at the top, followed by four input fields: First Name, Last Name, Email, and Password. Below the input fields is a blue button labeled REGISTER. The footer is a dark blue bar with the text Presidency University.

Figure 10 - Register Page



The screenshot shows the Login Page of the MERN Job Portal. The page has a blue header with the navigation menu: JOB PORTAL, HOME, REGISTER, and ATS. On the right side of the header, there are icons for a sun (theme toggle) and a user profile. The main content area is white and contains a login form. The form has a blue lock icon at the top, followed by two input fields: E-mail and Password. Below the input fields is a blue button labeled LOG IN. The footer is a dark blue bar with the text Presidency University.

Figure 11 - Login Page

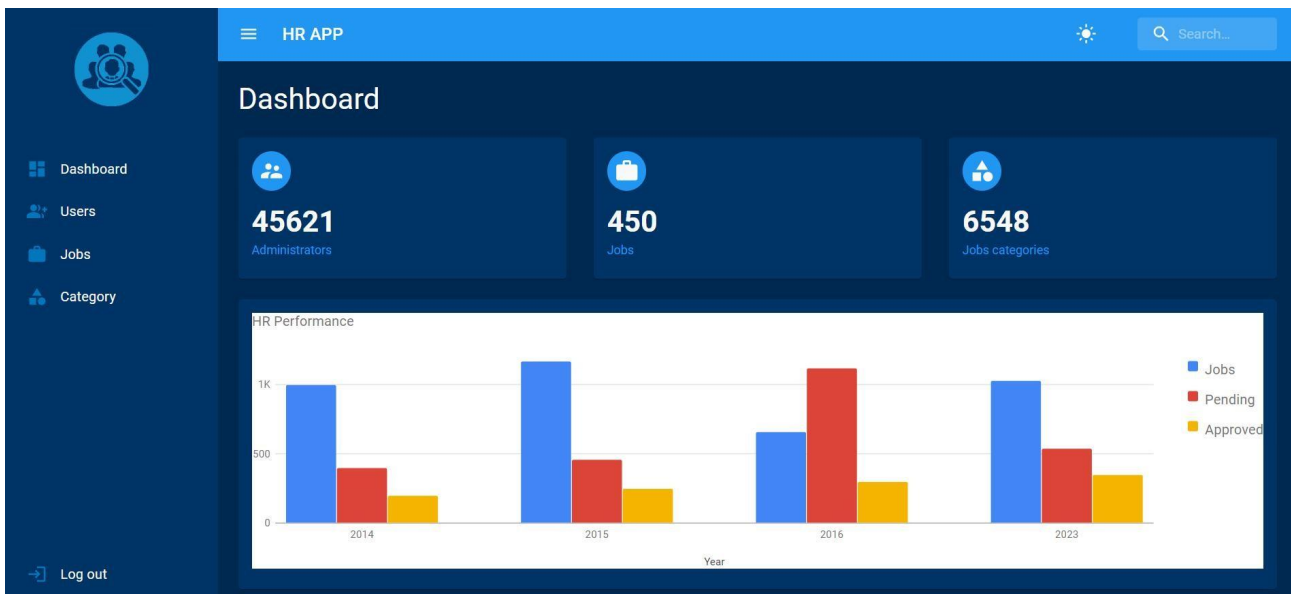


Figure 12 - Admin Dashboard

The All users page (HR APP) features a sidebar with navigation links: Dashboard, Users, Jobs, and Category. The main content area displays a table of users with columns: User ID, E_mail, User status, Creation date, and Actions. A + CREATE USER button is located in the top right corner.

User ID	E_mail	User status	Creation date	Actions
67519e939a42cc0...	aaa@gmail.com	Regular user	2024-12-05 18:12:03	EDIT DELETE
66f06f0811cb8b6...	akash@gmail.com	Regular user	2024-09-23 00:09:56	EDIT DELETE
66f06097d0267b4...	hem@gmail.com	Admin	2024-09-22 23:09:59	EDIT DELETE
66eed2f9040641...	somu@gmail.com	Regular user	2024-09-21 21:09:93	EDIT DELETE
66ee6e514880f54...	hemanth.725990...	Regular user	2024-09-21 12:09:42	EDIT DELETE

Rows per page: 100 1-9 of 9

Figure 13 - All Users

The screenshot shows the 'Register a Job' form in the HR APP. The form is centered on a dark blue background. It has a white border and contains the following fields:

- Title: A text input field with the placeholder 'Title'.
- Description: A text input field with the placeholder 'Description'.
- Salary: A text input field with the placeholder 'Salary'.
- Location: A text input field with the placeholder 'Location'.
- Category: A dropdown menu with a downward arrow.
- CREATE JOB: A blue button at the bottom of the form.

The left sidebar contains a menu with the following items: Dashboard, Users, Jobs, Category, and Log out. The top header bar is blue and contains the text 'HR APP' and a search bar.

Figure 14 - Register a job

The screenshot shows the 'Jobs list' page in the HR APP. The page has a dark blue background. At the top, there is a blue header bar with the text 'HR APP' and a search bar. Below the header, there is a green button labeled '+ CREATE JOB'. The main content area contains a table with the following columns:

☐	Job ID	Job name	Category	User	available	Salary	Actions
☐	6755b131f19d474...	hr	cloud	hemanth	Yes	\$456699	<button>EDIT</button> <button>DELETE</button>
☐	67519ef89a42cc0...	cloud engineer	frontend	hemanth	Yes	\$450000	<button>EDIT</button> <button>DELETE</button>
☐	673c6142060adf4...	Backend Engg	Backend	hemanth	Yes	\$1256596	<button>EDIT</button> <button>DELETE</button>
☐	673c5f4e060adf4f...	frontend developer	frontend	hemanth	Yes	\$40000	<button>EDIT</button> <button>DELETE</button>

At the bottom of the table, there is a pagination bar that reads 'Rows per page: 100' and '1-4 of 4'.

Figure 15 - Jobs List

HR APP

Search...

Create a Category

category

category name

CREATE CATEGORY

Dashboard

Users

Jobs

Category

Log out

Figure 16 - Create Job Category

HR APP

Search...

Jobs category

+ CREATE CATEGORY

☐	Category ID	Category	Create At	Actions
☐	6755b14df19d474...	business	2024-12-08 20:12:96	<button>EDIT</button> <button>DELETE</button>
☐	67519f0a9a42cc0...	cloud	2024-12-05 18:12:77	<button>EDIT</button> <button>DELETE</button>
☐	673c6113060adf4...	Backend	2024-11-19 15:11:72	<button>EDIT</button> <button>DELETE</button>
☐	673c5f42060adf4f...	frontend	2024-11-19 15:11:78	<button>EDIT</button> <button>DELETE</button>

Rows per page: 100 1-4 of 4

Dashboard

Users

Jobs

Category

Log out

Figure 17 - Job Categories

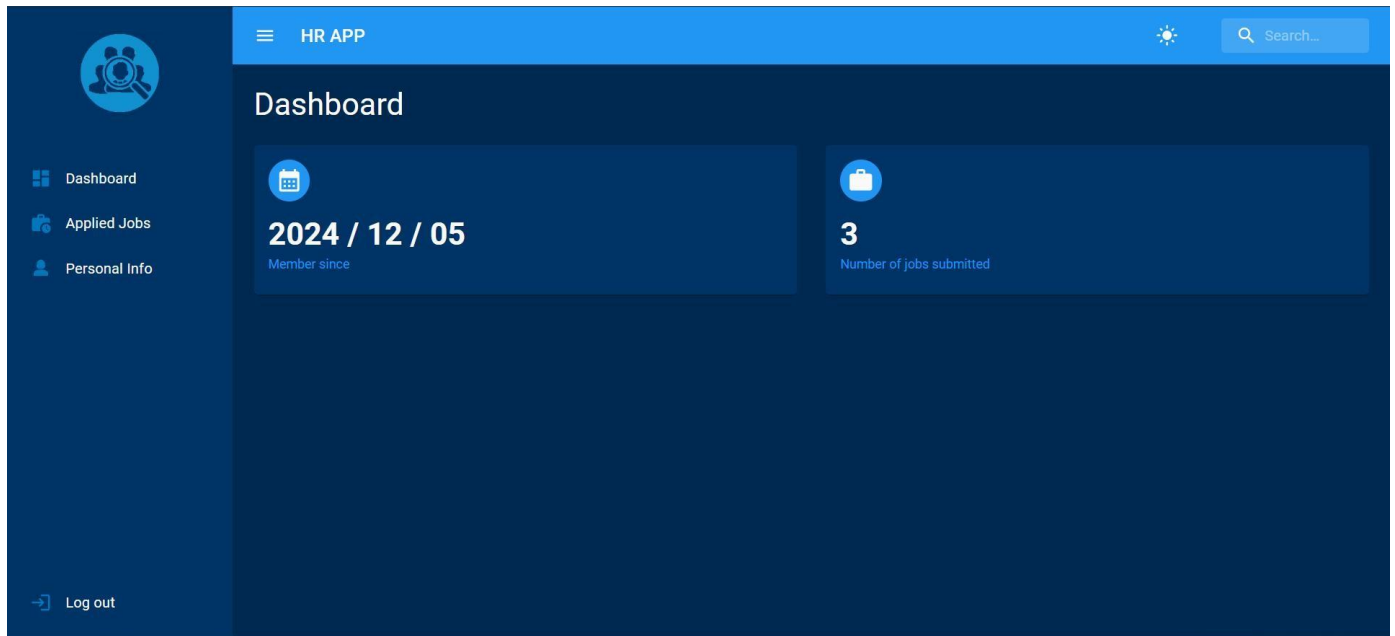


Figure 18 - Employee Dashboard

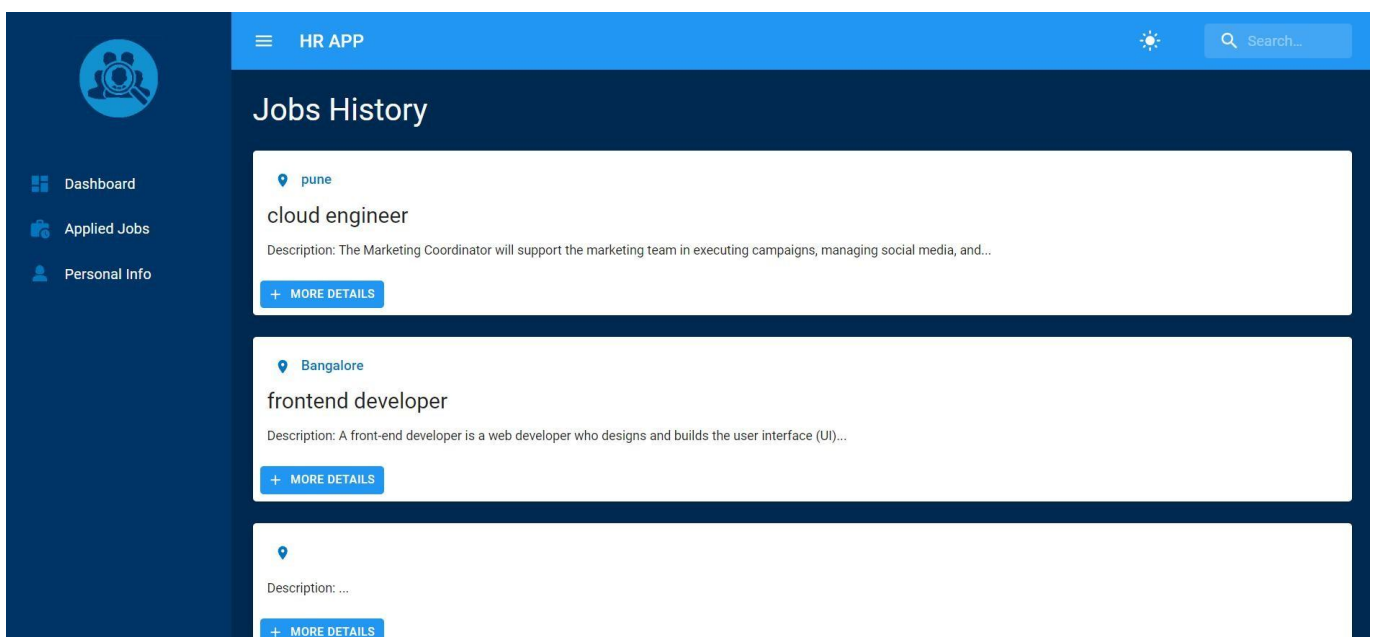


Figure 19 - Employee Jobs History

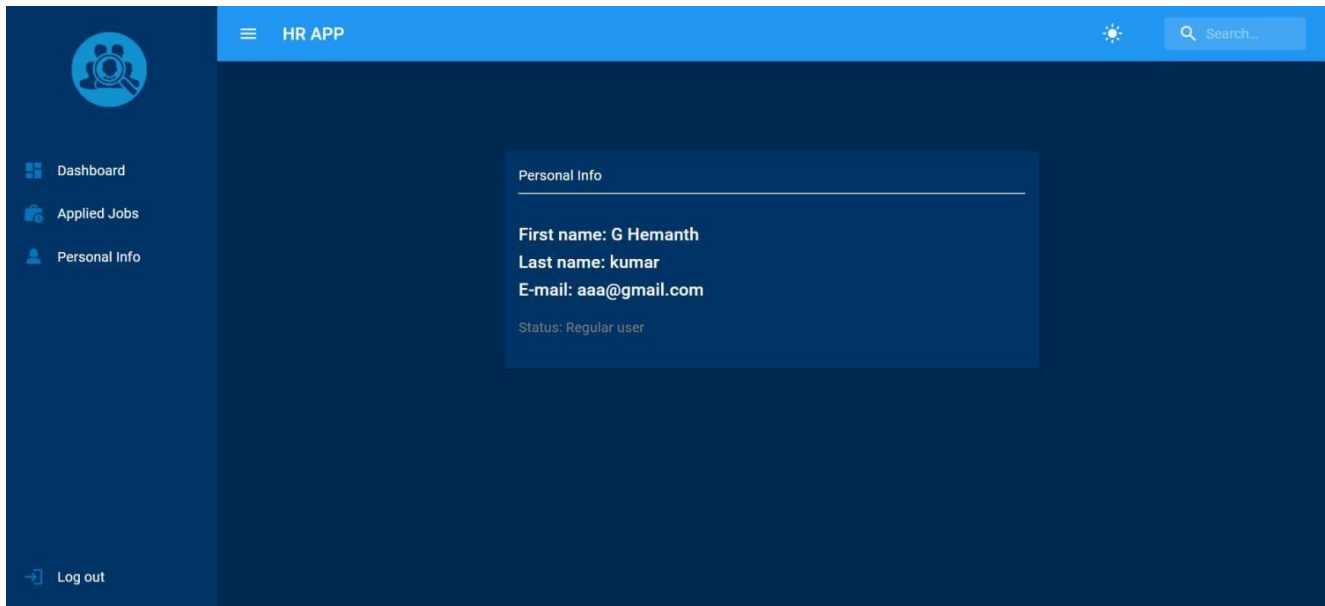


Figure 20 - Employee Personal Info

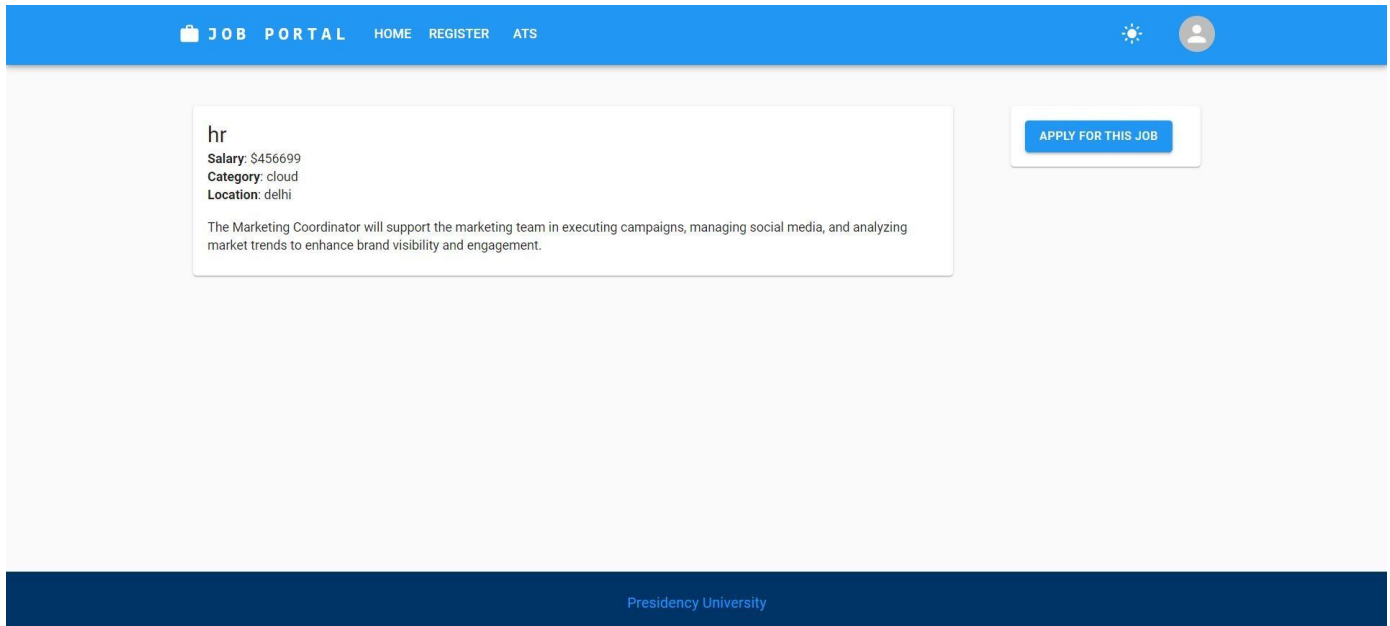
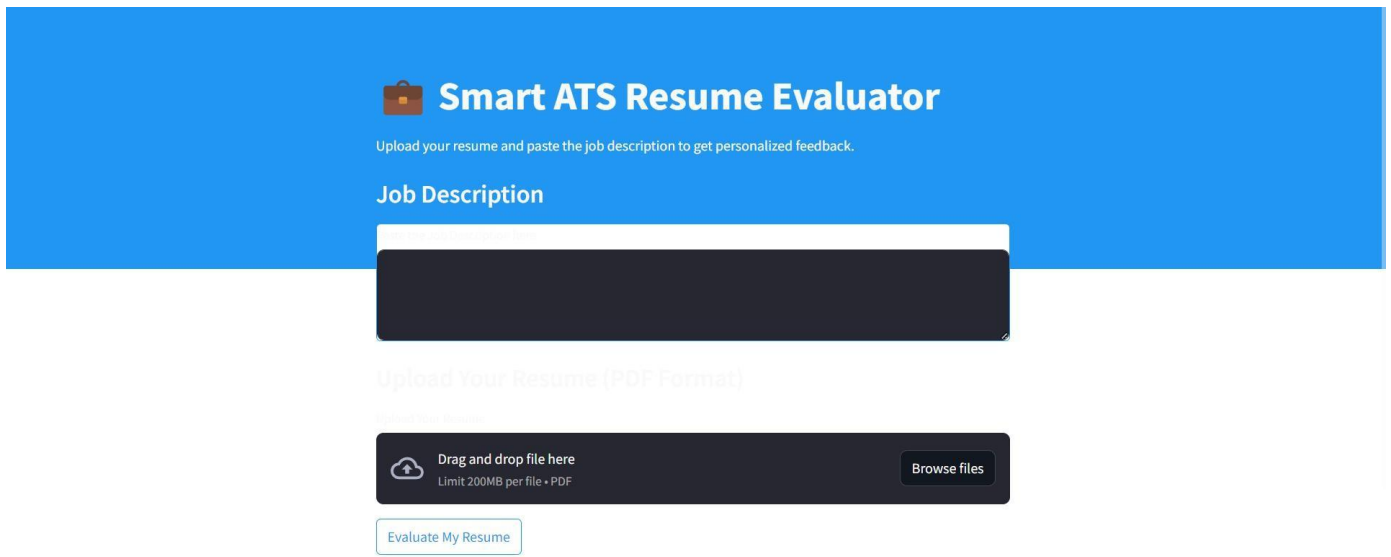


Figure 21 - Apply for Job



The interface features a blue header with a brown briefcase icon and the title "Smart ATS Resume Evaluator". Below the header, a message states: "Upload your resume and paste the job description to get personalized feedback." The "Job Description" section contains a dark grey text area. The "Upload Your Resume (PDF Format)" section includes a dark grey upload box with a cloud icon, the text "Drag and drop file here", "Limit 200MB per file • PDF", and a "Browse files" button. Below this is an "Evaluate My Resume" button.

Smart ATS Resume Evaluator

Upload your resume and paste the job description to get personalized feedback.

Job Description

Upload Your Resume (PDF Format)

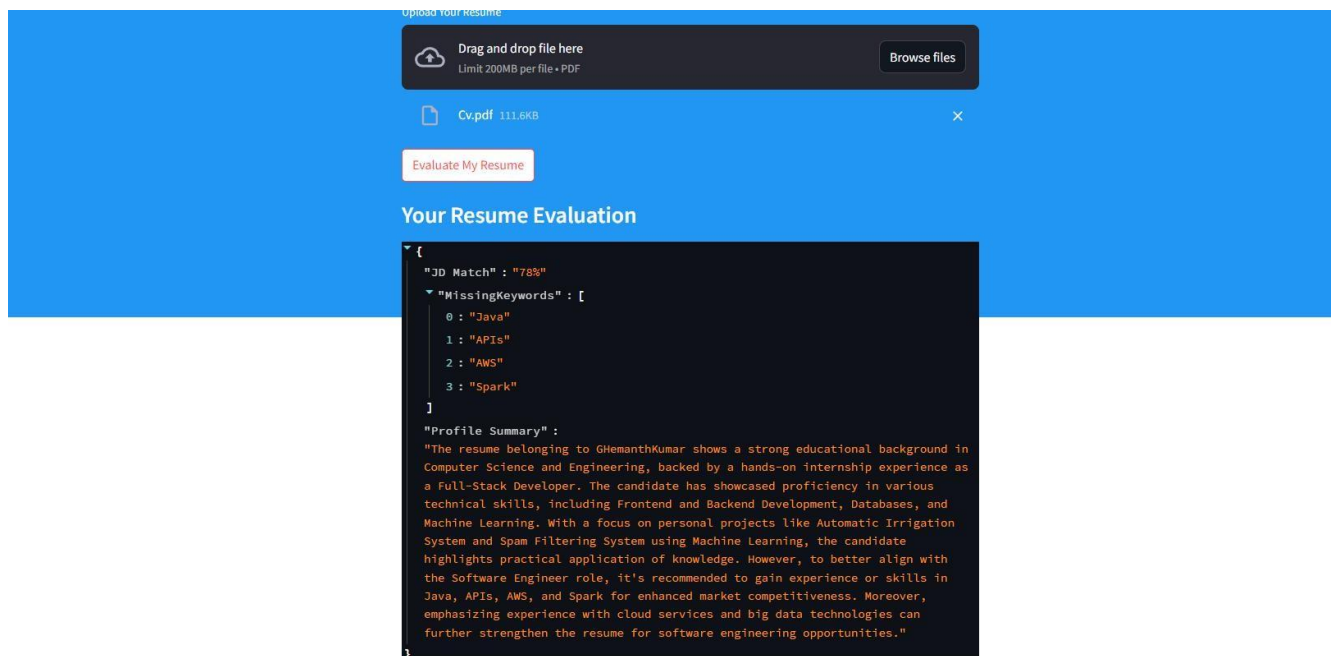
Upload Your Resume

 Drag and drop file here
Limit 200MB per file • PDF

Browse files

Evaluate My Resume

Figure 22 - Ats Evaluator



The interface shows the "Your Resume Evaluation" results. It includes the same upload section as Figure 22, but with a file "Cv.pdf" (111.6KB) uploaded. The "Evaluate My Resume" button is now orange. The evaluation results are displayed in a dark grey box with a light blue border. The results include a "JD Match" of 78%, a list of "MissingKeywords" (Java, APIs, AWS, Spark), and a "Profile Summary" paragraph.

Your Resume Evaluation

 Drag and drop file here
Limit 200MB per file • PDF

Browse files

 Cv.pdf 111.6KB ×

Evaluate My Resume

```
{
  "JD Match": "78%",
  "MissingKeywords": [
    0: "Java"
    1: "APIs"
    2: "AWS"
    3: "Spark"
  ]
}

"Profile Summary":
"The resume belonging to GHemanthKumar shows a strong educational background in Computer Science and Engineering, backed by a hands-on internship experience as a Full-Stack Developer. The candidate has showcased proficiency in various technical skills, including Frontend and Backend Development, Databases, and Machine Learning. With a focus on personal projects like Automatic Irrigation System and Spam Filtering System using Machine Learning, the candidate highlights practical application of knowledge. However, to better align with the Software Engineer role, it's recommended to gain experience or skills in Java, APIs, AWS, and Spark for enhanced market competitiveness. Moreover, emphasizing experience with cloud services and big data technologies can further strengthen the resume for software engineering opportunities."
```

Figure 23 - Evaluated Resume

APPENDIX-C

Similarity Index

Smart Hiring: MERN Job Portal with AI-Driven ATS Resume Optimization

ORIGINALITY REPORT

4 %	4 %	1 %	1 %
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

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G Hemanth Kumar

From

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Sustainable Development Goals

Goal 4: Quality Education

This project directly contributes to the promotion of lifelong learning opportunities by enabling job seekers to identify the gaps in their skills and qualifications through detailed feedback on their resumes. By guiding candidates on how to align their profiles with specific job descriptions, the system indirectly motivates them to pursue further education, certifications, or skill-building initiatives. This fosters a culture of continuous learning, ensures equal access to opportunities for skill development, and supports individuals in staying competitive in an ever-changing job market.

Goal 5: Gender Equality

The use of AI-driven evaluations in this project minimizes the unconscious biases often found in traditional hiring practices. By ensuring that resumes are assessed on merit and relevance rather than gender or other subjective factors, the system promotes fairness and inclusivity. This approach empowers women and other underrepresented groups to compete equally for job opportunities, contributing to gender parity in the workforce and fostering diverse and inclusive organizational cultures.

Goal 8: Decent Work and Economic Growth

By streamlining and optimizing the recruitment process, the project supports efficient job matching, reducing unemployment and underemployment rates. Its ability to provide constructive feedback enables job seekers to improve their qualifications and secure positions that align with their skills and aspirations. This enhances job satisfaction and productivity while ensuring sustainable economic growth through the development of a skilled and motivated workforce.

Goal 9: Industry, Innovation, and Infrastructure

This project leverages cutting-edge technologies such as artificial intelligence and natural language processing to modernize recruitment processes. By creating a robust digital infrastructure, it addresses the inefficiencies of traditional hiring methods and fosters innovation in workforce management. The system serves as a model for integrating

advanced technology into industry practices, enabling businesses to adapt to evolving challenges and thrive in competitive markets.

Goal 10: Reduced Inequalities

The system prioritizes skills and capabilities over traditional indicators such as educational pedigree or geographic location, ensuring that candidates from diverse backgrounds have equitable access to job opportunities. By reducing systemic biases and barriers in hiring, it fosters inclusion and empowers underrepresented groups, helping to create a more balanced and fair labor market. This contributes to reducing socioeconomic inequalities and building a more inclusive global workforce.

Goal 12: Responsible Consumption and Production

The project enhances sustainability in recruitment by digitizing processes and reducing the reliance on physical resources such as paper and manual effort. By automating repetitive tasks and optimizing resource usage, it contributes to more responsible consumption and production patterns in workforce management. This sustainable approach benefits both organizations and the environment, ensuring efficient and eco-friendly hiring practices.

Goal 17: Partnerships for the Goals

The system encourages collaboration between educational institutions, employers, and technology platforms to bridge skill gaps and design better training programs. By aligning the needs of the job market with educational curricula and fostering data-driven decision-making, the project builds partnerships that enhance the overall effectiveness of recruitment processes. These collaborations strengthen global efforts toward sustainable workforce development and economic progress.